



Dwight Look College of

ENGINEERING
TEXAS A&M UNIVERSITY

ECEN 404 Final Presentation

Team 24: Pyro Sensor

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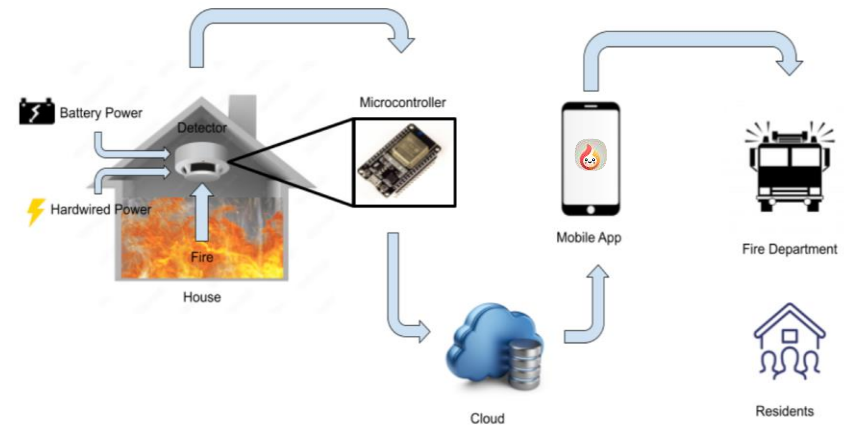
Oscar Hernandez

Sponsor: Rhett Guthrie

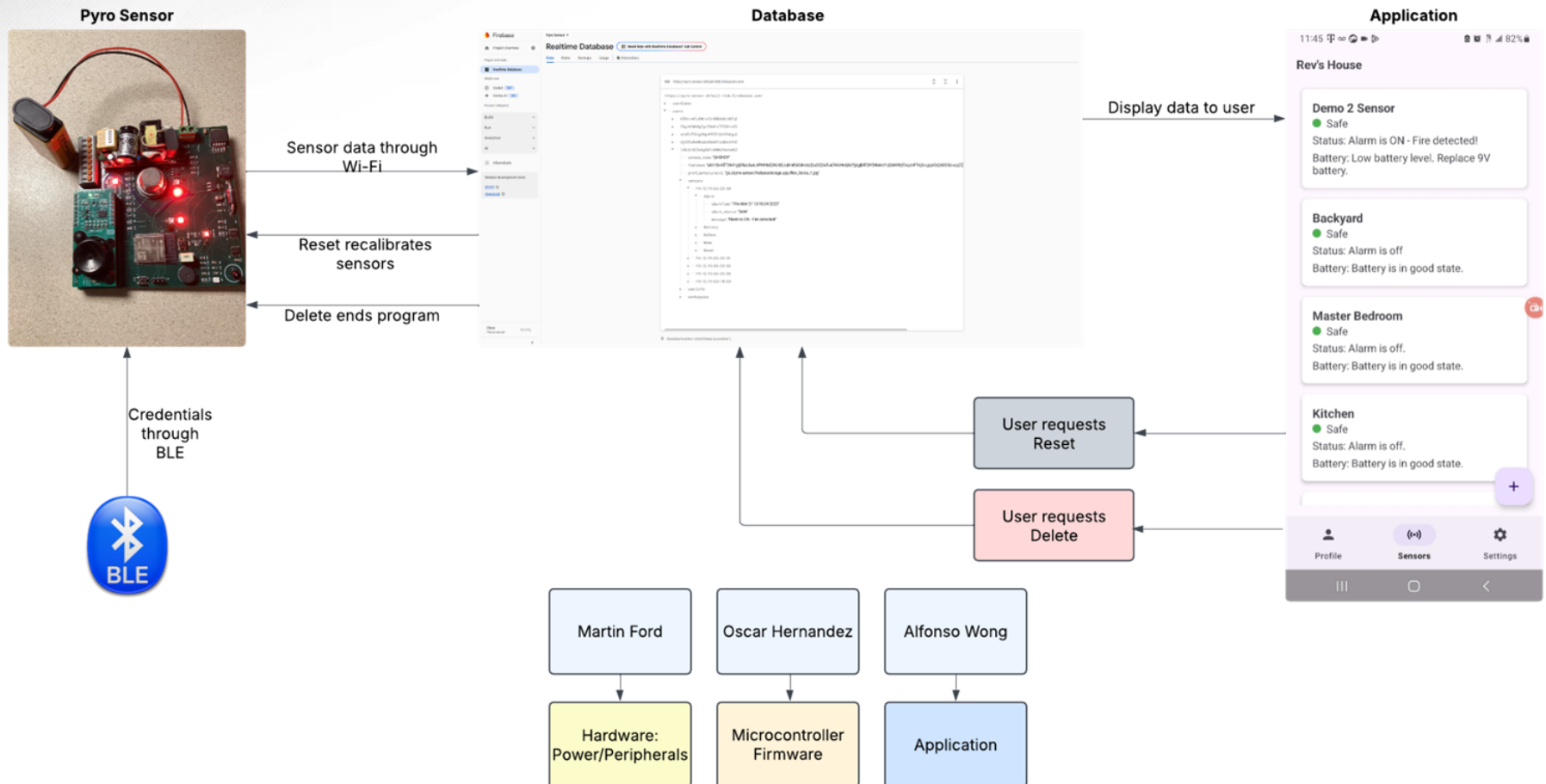
TA: Niloofar Borzooei

Problem Overview

- Problem statement:
 - Modern smoke detectors do not directly notify the emergency responders.
 - Fire department has limited data of residents in buildings.
- Our Pyro Sensor will:
 - Notify any user and emergency responders instantly through a mobile app of any incident.
 - Give fire department information about residents and pets.



Integrated Project Diagram



Hardware–Embedded System

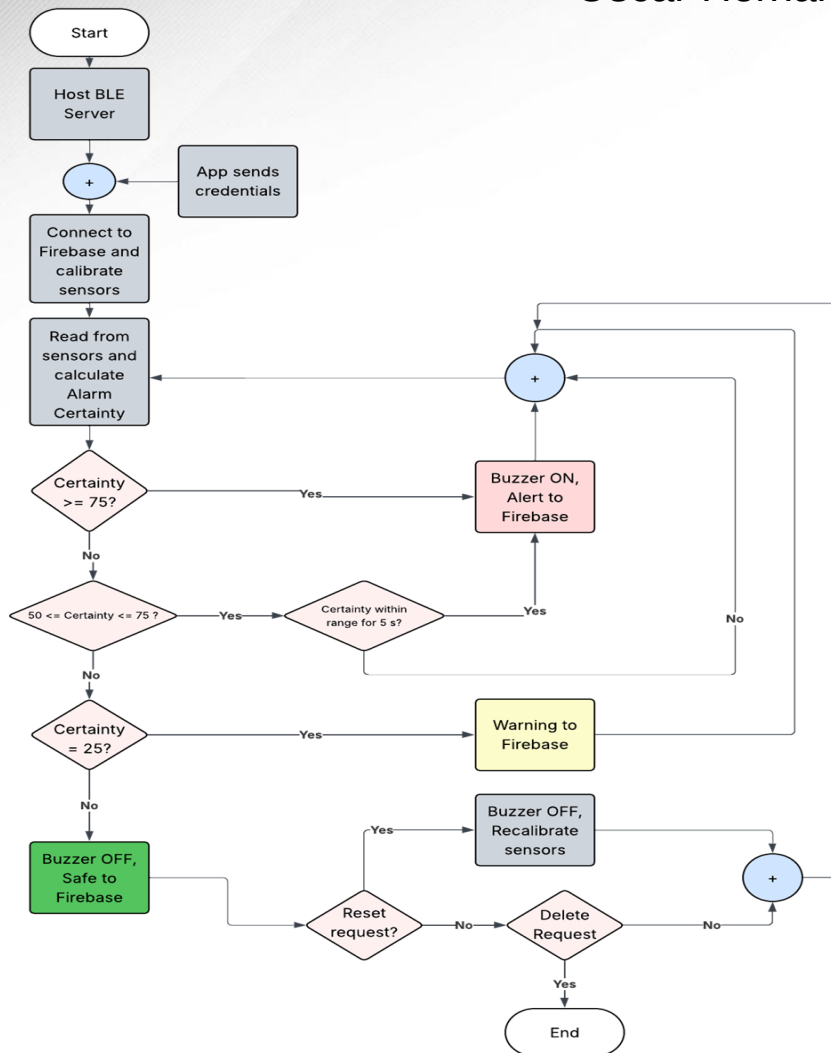
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Signal Processing

- Sensors detecting fire send signals to MCU.
- MCU uses algorithm with sensor hierarchy to minimize false alarms and maximize accuracy.
- MCU continuously monitors signals real-time status.
- MCU sends signal to toggle speaker alarm based on current status.

System Power

- MCU monitors battery power at all times and calculates battery level.
- MCU changes to idle or active mode (determining utilization of WIFI) to minimize power consumption and maximize battery life.



Microcontroller–App Connectivity and Integration

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Cloud Storage

- Use of Google service Firebase
 - Realtime Database - (Both MC and App use)
 - Authentication
 - Functions
 - Messages
 - Storage
- Both application and MC read and write to the database
- MC updates database and application displays updates

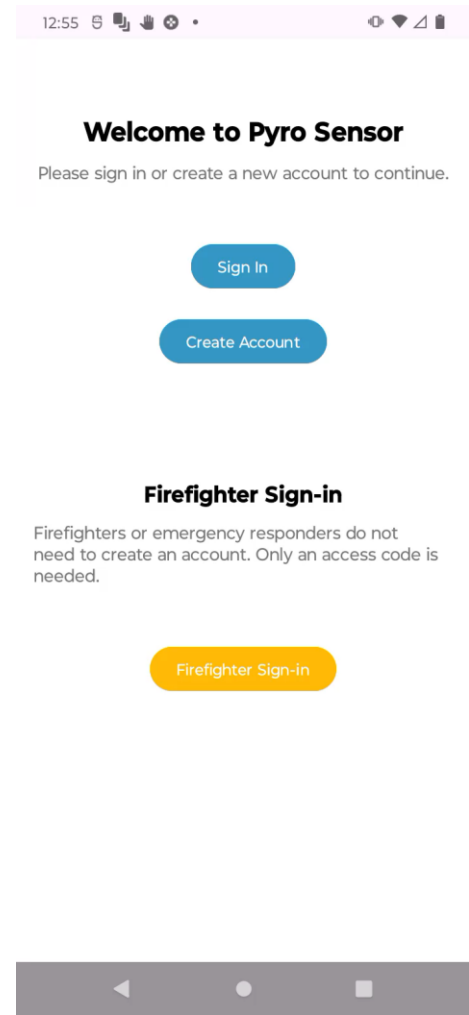
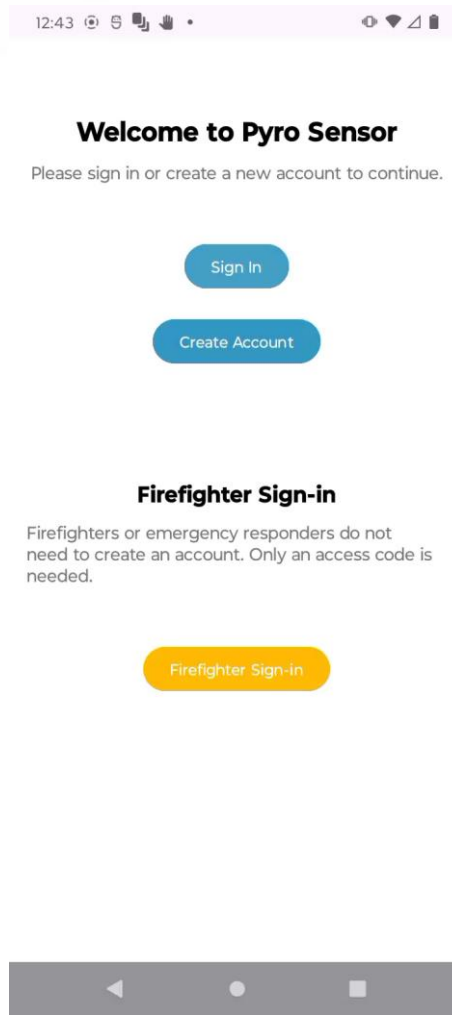
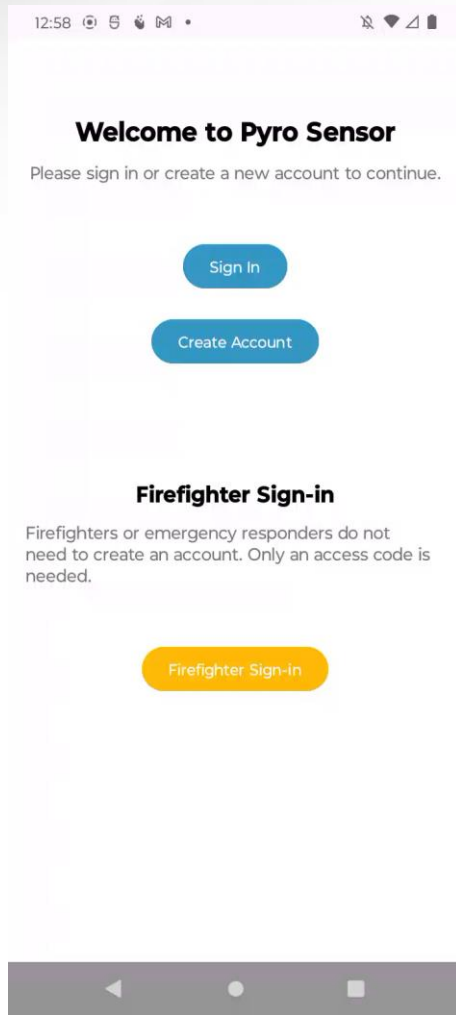
Application

- Sensor status can be seen by users as well as emergency responders through the application
- Users will get notification if a fire has been detected
- Emergency responders will get a code to access the application
- GUI is programmed in Kotlin through Android Studio



Microcontroller–App Connectivity and Integration

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Full-System Workflow and Operation

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Use Case: Fire Detection to User Notification

When smoke is detected by the sensor, the microcontroller processes the signal and sends a notification to the user's phone via Firebase. This was tested 15 times under various conditions.

- **Avg. detection-to-notification time: ~2.8 seconds**
- **Notification success rate: 100%**
- **App status:** Works whether open or closed

Use Case: Firefighter Access via Access Code

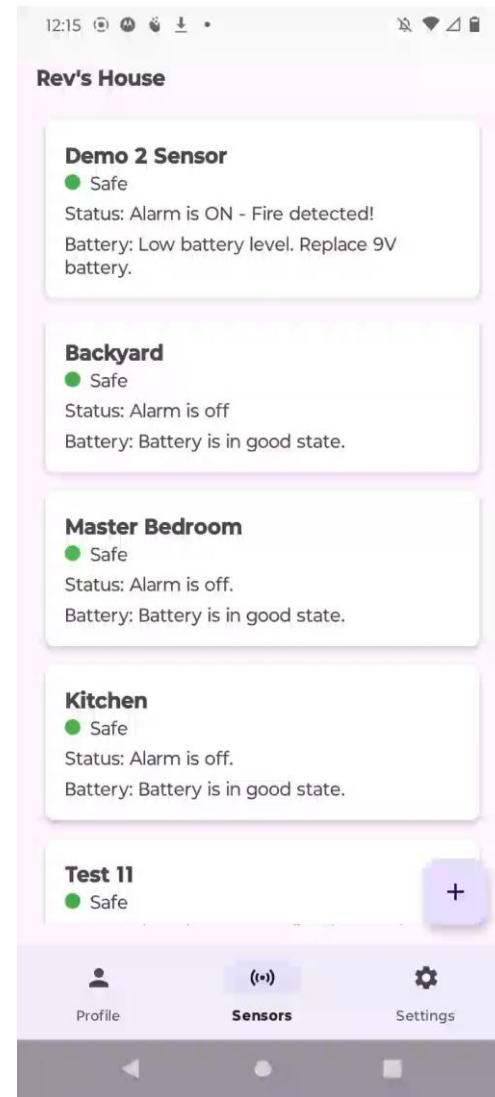
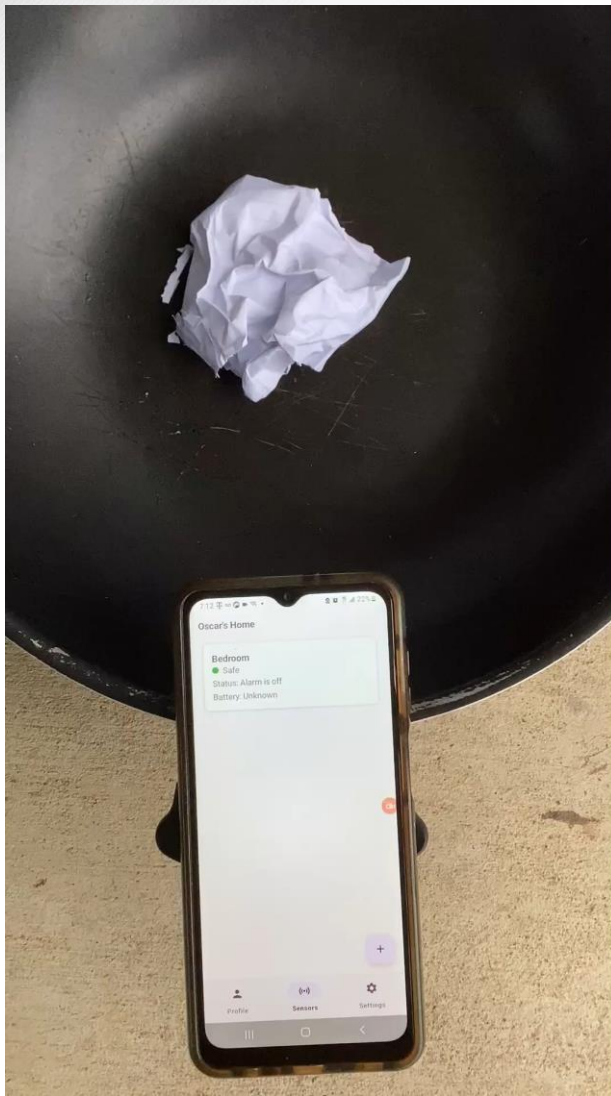
Once a workspace is created, a unique access code is generated and stored. In the event of a fire, this code is used by firefighters to retrieve building and occupant information from the mobile app.

- **Avg. data retrieval time: ~0.9 seconds**
- **Access success rate: 100%**



Full-System Workflow and Operation

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Thank you for your attention!

Questions?